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Project 2

1 Analytic part

a)

We argue that we can describe the flow using mass and momentum balance as

$$\partial_x v_x + \partial_y v_y + \partial_z v_z = 0 \tag{1}$$
$$\rho\left(\partial_t v_x + v_x \partial_x v_x + v_y \partial_y v_x + v_z \partial_z v_x\right) = -\partial_x p + \mu \nabla^2 v_x \tag{2}$$
$$\rho\left(\partial_t v_y + v_x \partial_x v_y + v_y \partial_y v_y + v_z \partial_z v_y\right) = -\partial_y p + \mu \nabla^2 v_y \tag{3}$$
$$\rho\left(\partial_t v_z + v_x \partial_x v_z + v_y \partial_y v_z + v_z \partial_z v_z\right) = -\partial_z p + \mu \nabla^2 v_z - \rho g \tag{4}$$

The first equation tells us that the fluid has no divergence as is therefore incompressible. For the other three, we can use momentum balance as a starting point:

$$\rho \frac{Dv}{dt} = f + \nabla \cdot T$$

where we know that the material derivative and stress tensor are:

$$\frac{Dv}{dt} = \partial_t v + (v \cdot \nabla) v$$
$$T = -pI + \lambda(\nabla \cdot v)I + 2\mu D$$

for our case, our fluid is Newtonian and therefore $\lambda = 0$. The divergence of the stress tensor becomes:

$$\nabla \cdot T = -p(\nabla \cdot I) + 2\mu(\nabla \cdot D) = -\nabla p + 2\mu(\nabla \cdot D)$$

D is the deformation tensor defined as:

$$D_{ij} = \frac{1}{2}\left(\partial_j v_i + \partial_i v_j\right)$$

where i and j are indices representing our coordinates. Looking at each direction $i = x, y, z$ separately:

$$\begin{aligned}(\nabla \cdot T)_i &= -\partial_i p + \mu \sum_j \partial_j (\partial_j v_i + \partial_i v_j) \\ &= -\partial_i p + \mu \left(\sum_j \partial_j \partial_j v_i + \sum_j \partial_j \partial_i v_j \right)\end{aligned}$$

The term $\sum_j \partial_j \partial_j v_i$ is equal to the Laplacian of the velocity $\nabla^2 v_i$, while the term $\sum_j \partial_j \partial_i v_j$ is zero because the divergence $\sum_j \partial_j v_j = 0$ since our fluid is incompressible. Therefore:

$$(\nabla \cdot T)_i = -\partial_i p + \mu \nabla^2 v_i$$

Coming now back to our momentum balance relation, we can write an expression for each of the directions x, y, and z separately by writing them with the index i:

$$\rho \left(\partial_t v_i + \sum_j v_j \partial_j v_i \right) = f_i - \partial_i p + \mu \nabla^2 v_i$$

Replacing i with each cartesian coordinate at a time gives the bottom three equations. For x and y, the volume forces in that direction are zero and the f term drops off, however for z this term is equal to $f = -\rho g$ due to gravity.

b)

In our situation, the equations, using $v_x = u$, simplify to

$$\rho \partial_t u = -\partial_x p + \mu\, \partial_z^2 u, \tag{5}$$
$$\partial_z v_z = 0, \qquad \text{hence } v_z = 0, \tag{6}$$
$$0 = -\partial_y p, \tag{7}$$
$$0 = -\partial_z p - \rho g, \tag{8}$$
$$p(x,z,t) = -\rho g z + C(x,y,t). \tag{9}$$

We can do this, because we know the velocity field $\vec{v} = (u(z,t), 0, 0)^T$. $v_y, v_z = 0$, and therefore their derivatives will be 0 too. v_x only depends on z and t, and therefore also $\partial_x v_x, \partial_y v_x = 0$. This allows us to do the following simplifications.

Equation 2 simplifies to equation 5:

$$\begin{aligned} \rho\left(\partial_t v_x + v_x \partial_x v_x + v_y \partial_y v_x + v_z \partial_z v_x\right) &= -\partial_x p + \mu \nabla^2 v_x \\ \implies \quad \rho\left(\partial_t v_x + 0 + 0 + 0\right) &= -\partial_x p + \mu \partial_z^2 v_x \\ \implies \quad \rho\, \partial_t u &= -\partial_x p + \mu\, \partial_z^2 u \end{aligned}$$

Equation 6 is trivial from the known velocity field $\vec{v} = (u(z,t), 0, 0)^T$.

Equation 7 is a simplified form from 3:

$$\begin{aligned} \rho\left(\partial_t v_y + v_x \partial_x v_y + v_y \partial_y v_y + v_z \partial_z v_y\right) &= -\partial_y p + \mu \nabla^2 v_y \\ \implies \quad \rho\left(0 + 0 + 0 + 0\right) &= -\partial_y p + \mu \cdot 0 \\ \implies \quad 0 &= -\partial_y p \end{aligned}$$

Equation 8 comes from 4

$$\begin{aligned} \rho\left(\partial_t v_z + v_x \partial_x v_z + v_y \partial_y v_z + v_z \partial_z v_z\right) &= -\partial_z p + \mu \nabla^2 v_z - \rho g \\ \implies \quad \rho\left(0 + 0 + 0 + 0\right) &= -\partial_z p + \mu \cdot 0 - \rho g \\ \implies \quad 0 &= -\partial_z p - \rho g \end{aligned}$$

Equation 9 comes from taking the integral over equation 8 and knowing $\partial_y p = 0$

$$\begin{aligned} 0 &= -\partial_z p - \rho g \\ \partial_z p &= -\rho g \\ \int \partial_z p dz &= \int -\rho g dz \\ p(x,y,z,t) &= -\rho g z + C(x,y,t) \end{aligned}$$

We are assuming that the plate that is oscillating is infinite and uniform in the x, y-direction. Therefore, the velocity cannot depend on x, y. Also, initially, the pressure must be constant in the x, y directions, otherwise there would be a pressure gradient, giving the water a velocity in the x, y directions. For all t there is always a z where the fluid is still not moving. The pressure can still not depend on x, y. This means that $\partial_x p = \partial_y p = 0$ and, therefore, C cannot depend on x, y. Knowing this we can simplify equation 5.

$$\rho\, \partial_t u = -\partial_x p + \mu\, \partial_z^2 u \implies \rho \partial_t u = \mu\, \partial_z^2 u$$

c)

The equations of motion and boundary conditions then become

$$\begin{aligned}
&(1) \quad \rho\,\partial_t u = \mu\,\partial_z^2 u,\\
&(2) \quad z = 0:\ u = U(t),\\
&(3) \quad z = \infty:\ u \to 0,\\
&(4) \quad t = 0:\ u = 0.
\end{aligned}$$

This is beacause TODO

d)

We can find with BC (2) that the solution should have a form:

$$u(z,t) = f(z)\cdot U_0 \sin\left(2\pi\frac{t-\tau(z)}{T}\right),$$

with $f(0) = 1$ and $\tau(0) = 0$. Here, $f(z)$ is a function that describes how the amplitude of the oscillations changes with height and $\tau(z)$ is a function that describes how the phase of the oscillations changes with height. Since we know that the amplitude of oscilation is damped for higher z, we can expect that $f(z)$ is a decreasing function of z. Furtermore, Damping is usually exponential, which is why $f(z) = e^{-k}$ is a reasonable guess. The phase-shift is necessary, since we know that the oscilation only moves up the water with a ceratin delay. Then, the equaiton of motion becomes

$$u(z,t) = U_0 e^{-kz} \sin\left(2\pi\frac{t-\tau(z)}{T}\right),$$

where k must be a positive constant, since otherwise there would be no damping effect.

Note that we implicitly used BC (3), as this is determining the damping. Furthermore, this is only reasonable after a few oscilations, which is why we cannot apply BC (4). This leaves us with BC (1), with which we can determine k and $\tau(z)$:

$$\begin{aligned}
\rho\,\partial_t u &= \mu\,\partial_z^2 u\\
\rho U_0 e^{-kz}\frac{2\pi}{T}\cos\left(2\pi\frac{t-\tau(z)}{T}\right) &= \mu\partial_z\left(U_0 e^{-kz}(-k)\sin\left(2\pi\frac{t-\tau(z)}{T}\right) - U_0 e^{-kz}\cos\left(2\pi\frac{t-\tau(z)}{T}\right)\frac{2\pi\partial_z\tau(z)}{T}\right)\\
\frac{\rho}{\mu}e^{-kz}\frac{2\pi}{T}\cos\left(2\pi\frac{t-\tau(z)}{T}\right) &= e^{-kz}\left(k^2\sin\left(2\pi\frac{t-\tau(z)}{T}\right) + k\cos\left(2\pi\frac{t-\tau(z)}{T}\right)\frac{2\pi\partial_z\tau(z)}{T}\right)\\
&\quad - e^{-kz}\cos\left(2\pi\frac{t-\tau(z)}{T}\right)\frac{2\pi}{T}\left((-k\partial_z\tau(z)) + \partial_z^2\tau(z)\right)\\
&\quad - e^{-kz}\sin\left(2\pi\frac{t-\tau(z)}{T}\right)\left(\frac{2\pi\partial_z\tau(z)}{T}\right)^2\\
\frac{2\pi\rho}{\mu T}\cos\left(2\pi\frac{t-\tau(z)}{T}\right) &= \sin\left(2\pi\frac{t-\tau(z)}{T}\right)\left(k^2 - \left(\frac{2\pi\partial_z\tau(z)}{T}\right)^2\right)\\
&\quad + \cos\left(2\pi\frac{t-\tau(z)}{T}\right)\left(2k\frac{2\pi\partial_z\tau(z)}{T} - \left(\frac{2\pi\partial_z^2\tau(z)}{T}\right)\right)
\end{aligned}$$

Since we know that this must be true for all of t, we can find that the sin terms must be zero and the cos terms must be equal on both sides. This gives us the two equations:

$$\begin{aligned}
k^2 - \left(\frac{2\pi\partial_z\tau(z)}{T}\right)^2 &= 0,\\
2k\frac{2\pi\partial_z\tau(z)}{T} - \left(\frac{2\pi\partial_z^2\tau(z)}{T}\right) &= \frac{2\pi\rho}{\mu T}.
\end{aligned}$$

The first one reduces to:

$$k = \frac{2\pi\partial_z\tau(z)}{T} \Leftrightarrow \partial_z\tau(z) = \pm\frac{kT}{2\pi} \to \tau(z) = \pm\frac{kT}{2\pi}z + C$$

where we only take the positive value of τ because the particles higher up cannot be ahead of the particles below. To determine the constant, we use BC (2):[1]

$$\begin{aligned} u(0,t) &= U(t) \\ U_0 \sin\left(2\pi\frac{t-\tau(0)}{T}\right) &= U_0 \sin\left(2\pi\frac{t}{T}\right) \\ \Rightarrow \tau(0) = 0 &\Rightarrow \frac{kT}{2\pi}0 + C = 0 \Rightarrow C = 0 \end{aligned}$$

With the second equation, we can determine the value of k:

$$\begin{aligned} 2k\partial_z\tau(z) - \partial_z^2\tau(z) &= \frac{\rho}{\mu} \\ \Leftrightarrow \tau'' - 2k\tau' + \frac{\rho}{\mu} &= 0 \\ 0 - 2k\frac{kT}{2\pi} + \frac{\rho}{\mu} &= 0 \\ \Rightarrow k &= \sqrt{\frac{\rho 2\pi}{2\mu T}} = \sqrt{\frac{\rho\omega}{2\mu}} \end{aligned}$$

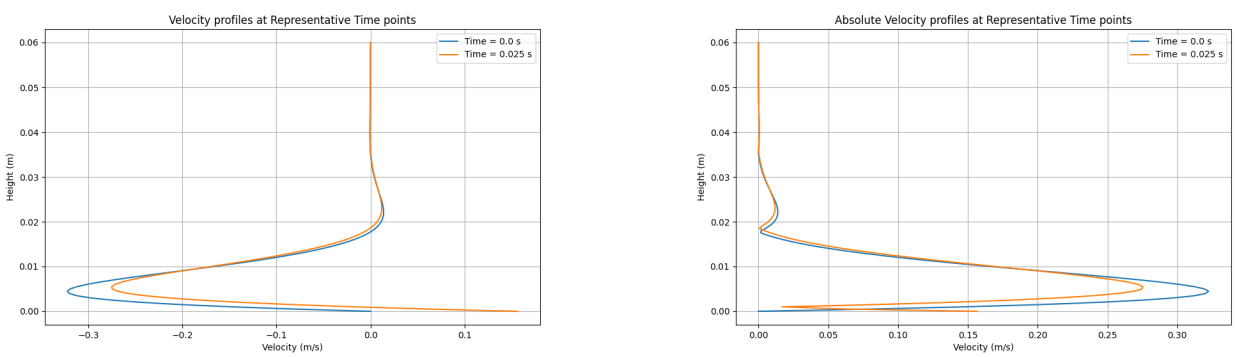
We find that $k = \sqrt{\frac{\rho\omega}{2\mu}}$ where $\omega = 2\pi/T$ and $\tau(z) = \frac{kT}{2\pi}z$.

The statement that k increases with ω is equivalent to saying that a faster oscilation will experience stronger damping. This statement is reasonable, since the acceleration of the next layer of fluid due to the friction is not instentanous. However, when now the oscilation is faster, the next layer of fluid will not have enough time to accelerate before the oscilation changes direction again, therefore the whole motion experiences stronger damping per height.

e)

The velocity field $u(z,t)$ changes over a period T as a damped sinusodial function along the z-axis.

When we plot the field at time $t = 0$ (Figure 1a) and $t = T/4$ (Figure 1b) we see that TODO



(a) (b)

Figure 1: Velocity field for $t = 0$ (a) and $t = T/4$ (b)

[1] Of course, used this already above, for completion the calculation is included here.

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2 Computational part

All explanations for the computational part are in the jupyter notebook.